

Persistent Homology of Music Network with Three Different Distances

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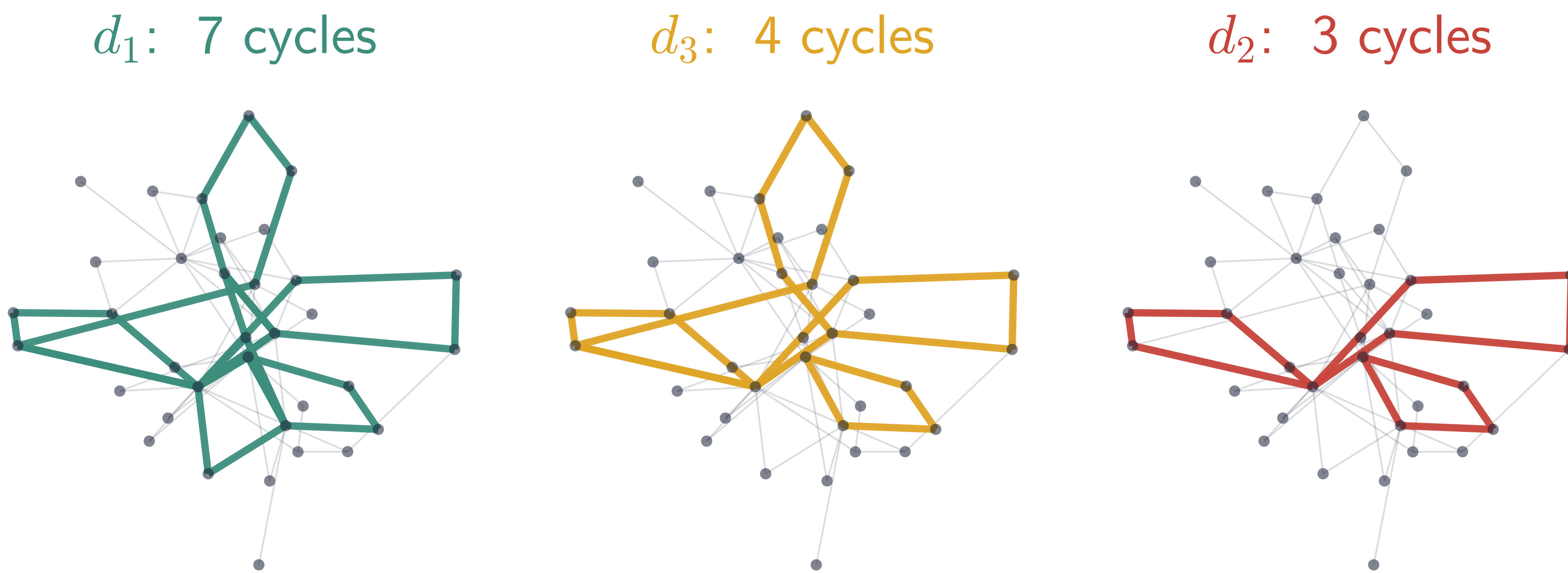
Project page

The big idea

Distance is a **tunable resolution knob** on musical structure — sweeping $d_1 \rightarrow d_3 \rightarrow d_2$ distills a piece down to its **core cyclic skeleton**.

- **Three path distances:** d_1 (min edges), d_2 (min weight — the only metric), d_3 (hybrid).
- **Proven ordering** $d_2 \leq d_3 \leq d_1$ lifts to a **nontrivial barcode injection** $\text{bcd}_1(d_2) \hookrightarrow \text{bcd}_1(d_3) \hookrightarrow \text{bcd}_1(d_1)$ (Heo–Choi–Jung 2025).
- **Validated** on real Korean court music — an exact match to the paper’s Table 1.

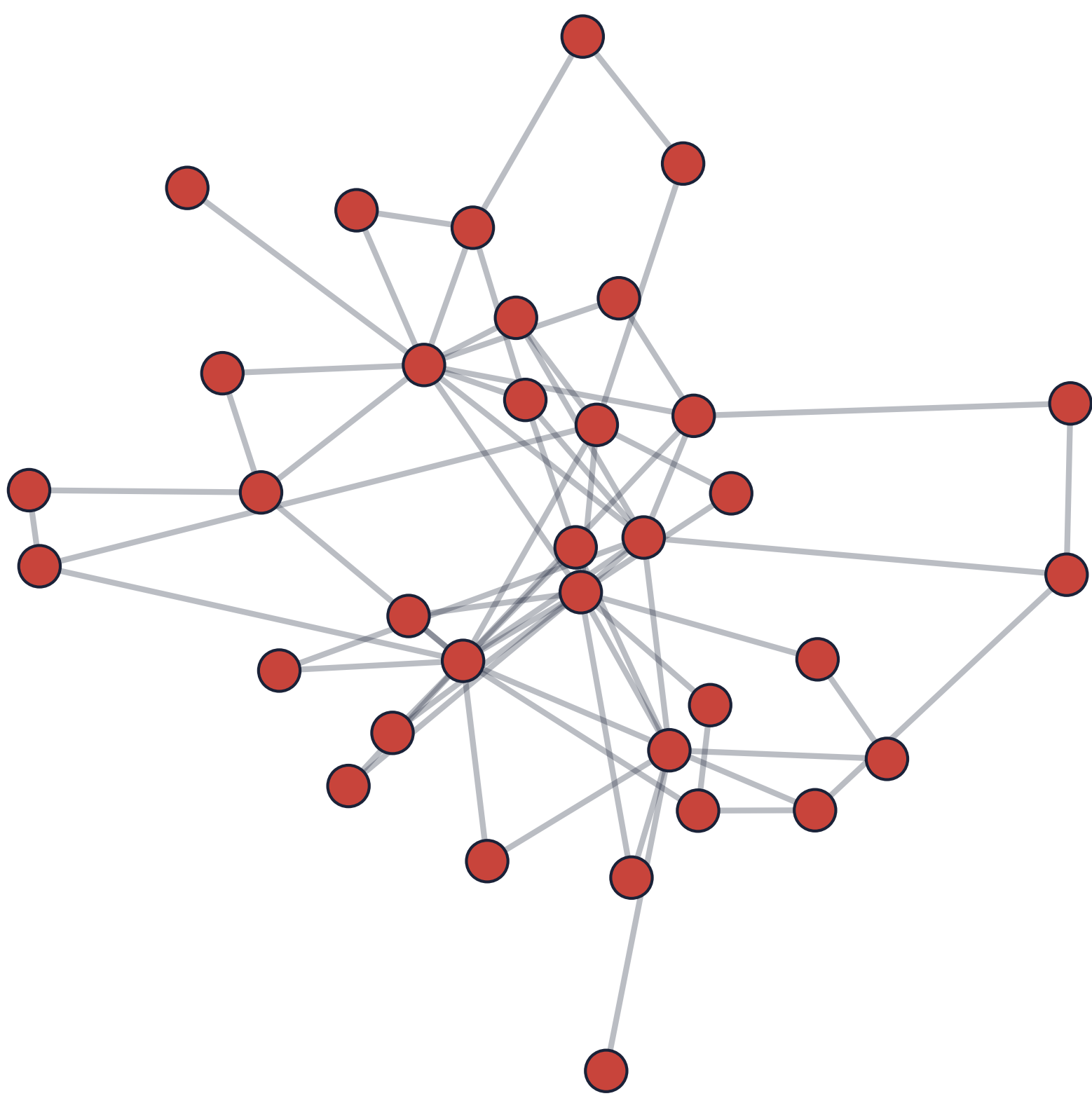
Cycle skeleton extraction



On *Sanghyeondodeuri* (geomungo): as $d_1 \rightarrow d_3 \rightarrow d_2$, the H_1 barcode thins $|\text{bcd}_1| = 7 \rightarrow 4 \rightarrow 3$ — only the robust core loops survive.

From music to a graph

Each **node** = (pitch, duration); an **edge** joins consecutive notes; the **weight** is $1/(\text{co-occurrence count})$ — frequent neighbors are closer. Rests are skipped.



The music network of *Sanghyeondodeuri* (geomungo).

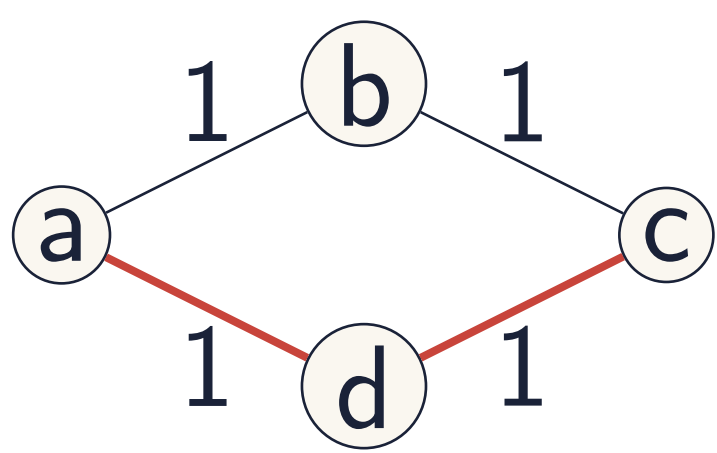
Three path distances

On the music graph $M = (V, E, W)$, a path P has **hop count** $\ell(P)$ and **weight** $\omega(P) = \sum_{e \in P} W(e)$; let $h = \min_P \ell(P)$ be the fewest hops $v \rightarrow w$.

d_1 $d_1(v, w) = \omega(P)$ along a **fewest-edge** path ($\ell(P) = h$).

d_2 $d_2(v, w) = \min_P \omega(P)$ — least total weight; *the only true metric*.

d_3 $d_3(v, w) = \min\{\omega(P) : \ell(P) = h\}$ — least weight *among* fewest-edge paths.



Path-based distance uses existing edges (red), not a new direct link.

Only d_2 is a metric

d_2 satisfies the triangle inequality; d_1 and d_3 can violate it (a heavy direct edge). Yet all three give meaningful musical structure.

A universal ordering

For every pair of nodes,

$$d_2(v, w) \leq d_3(v, w) \leq d_1(v, w).$$

This carries over to one-dimensional persistent homology.

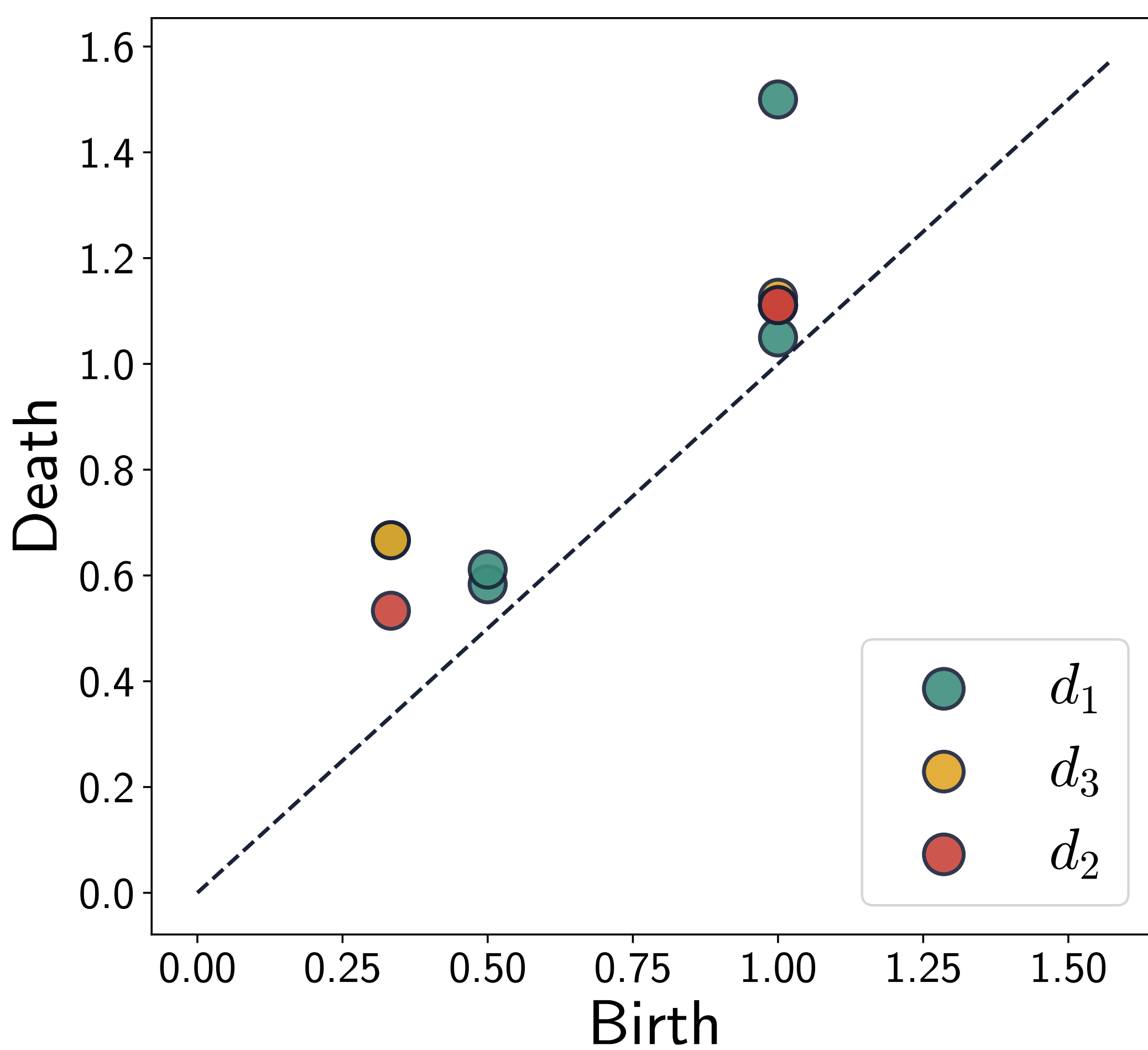
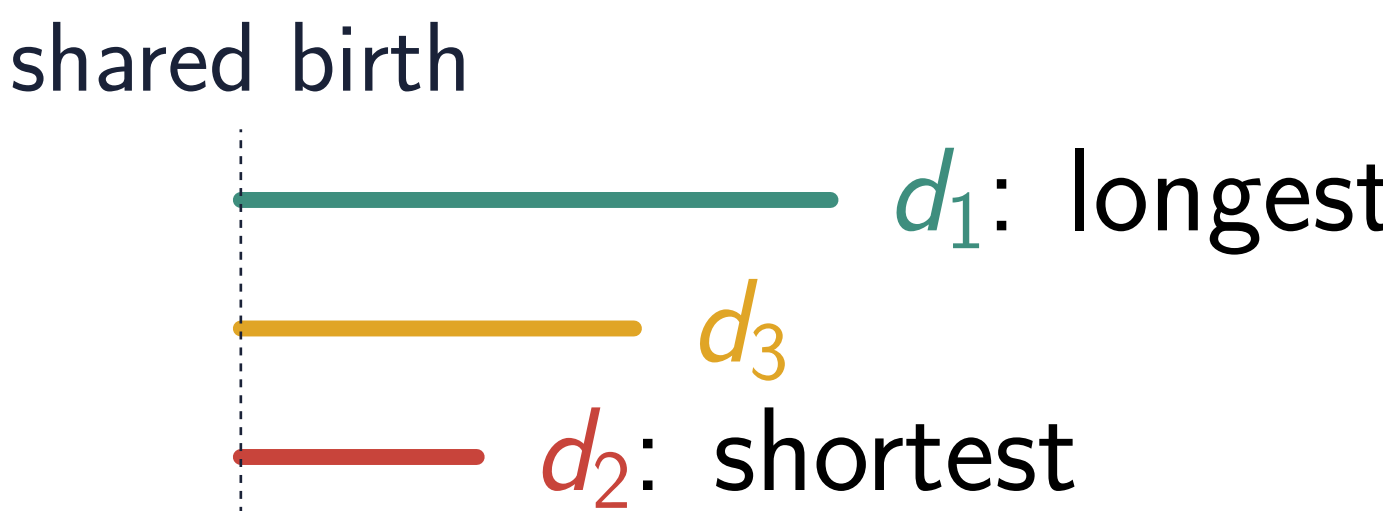
Barcode injection

The birth edges nest, $B_2 \subseteq B_3 \subseteq B_1$, giving injections

$$\text{bcd}_1(d_2) \hookrightarrow \text{bcd}_1(d_3) \hookrightarrow \text{bcd}_1(d_1).$$

This injection is not automatic: that an ordering of distances embeds the whole H_1 barcode is a **nontrivial theorem** for path-representable graph distances, proved in **Heo, Choi & Jung 2025** (arXiv:2501.03553).

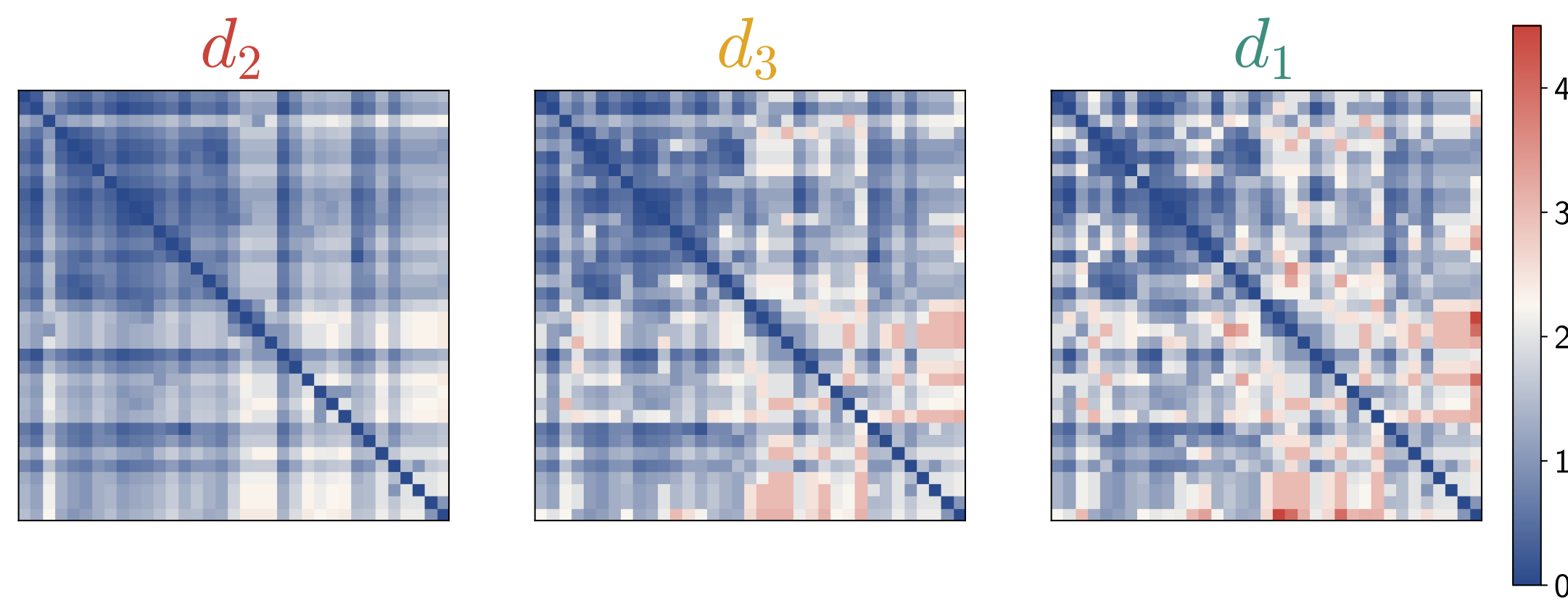
Each cycle keeps its **birth**; its **lifespan grows** $d_2 \leq d_3 \leq d_1$, and the number of cycles can only increase.



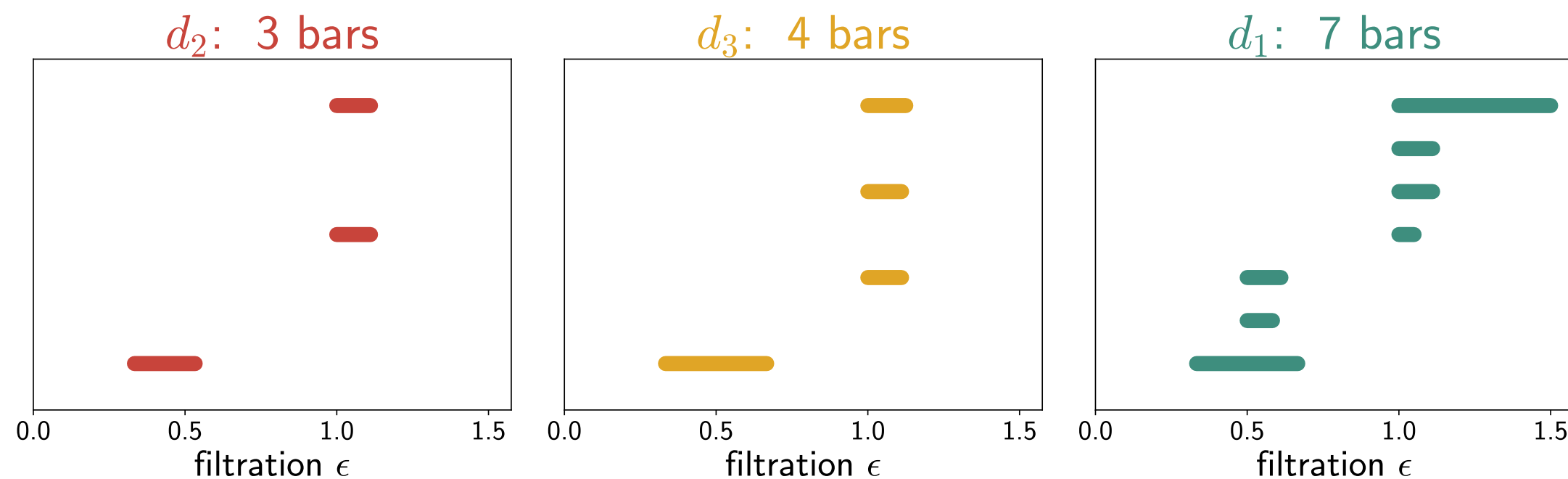
On real music: d_1 points lie above d_2 — longer lifespans, shared births.

Why it matters

Evidence on real music



Distance matrices: d_2 (cool/smooth) $\rightarrow d_1$ (hot) the ordering is visible.



H_1 barcodes: $|\text{bcd}_1(d_1)|, |\text{bcd}_1(d_3)|, |\text{bcd}_1(d_2)| = 7, 4, 3$; lifespans grow, births shared.

Distilling the core

Ranking d_1 cycles by how far down they survive ($\rightarrow d_3 \rightarrow d_2$) gives a principled **importance hierarchy of musical loops** — a multi-resolution descriptor of a piece.

Takeaways & future work

- Distance choice = a tunable lens; results are nested, not arbitrary.
- Toward **AI music composition** with multiple stylistic perspectives from one source.



Interactive explorer & audio demos on the project page.